PASSWORD-BASED DIGITAL DOOR LOCK SYSTEM

Using Verilog HDL: RTL Design, Simulation and Static Timing Analysis

Technology: SKY130 PDK

Project Report

Submitted by: _____

Department: _____

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Abstract

This project presents a password-based digital door lock designed using Verilog Hardware Description Language (HDL). The system checks a serial password entered by a user and controls an unlock output according to the received sequence. A correct password, 1011, is accepted, while an incorrect sequence such as 1001 is rejected. The design is verified through simulation and synthesized into a gate-level representation. Static Timing Analysis (STA) is performed using timing libraries associated with the SKY130 process design kit (PDK), considering typical, slow and fast operating corners. The project demonstrates key stages of a digital VLSI workflow, from RTL design and functional verification to synthesis and timing evaluation.

1. Introduction

Digital access-control systems compare an entered code with a predefined password before allowing access. Implementing the control logic in Verilog provides a practical introduction to sequential digital design, finite-state behaviour, simulation, synthesis and timing analysis. This project focuses on a compact password-controlled lock suitable for educational demonstration.

2. Objectives

- Design a synthesizable password-lock controller using Verilog HDL.
- Accept the correct serial password and reject an incorrect sequence.
- Verify the design using a Verilog testbench and waveform output.
- Synthesize the RTL and inspect the mapped design.
- Evaluate timing behaviour at typical, slow and fast PVT corners using SKY130 timing libraries.

3. Design Methodology

1. **RTL design:** describe the password checking and lock-control logic in Verilog.
2. **Testbench creation:** provide clock/reset and password input sequences.
3. **Functional simulation:** test valid and invalid passwords and inspect the waveform.
4. **Synthesis:** translate the RTL into a mapped gate-level netlist.
5. **Static Timing Analysis:** evaluate timing paths using available timing libraries and corner-specific scripts.
6. **Physical-design flow:** use the configured OpenLane/SKY130 flow where applicable and review the available implementation reports.

4. Functional Verification

Test case	Input password	Expected behaviour	Recorded outcome
Valid password	1011	Unlock accepted	Simulation passed
Invalid password	1001	Access rejected	Simulation passed

The testbench and generated VCD waveform are retained in the project repository. The results above reflect the recorded simulation tests; they do not constitute a security assessment of a production door-lock system.

5. Synthesis and Timing Analysis

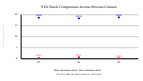
The RTL was synthesized into a mapped Verilog representation. STA was performed using the available corner-specific timing libraries and scripts. Slack values below are copied from the existing project logs; units and the exact timing constraints should be confirmed from the original

reports before making sign-off claims.

Timing corner	Maximum slack (reported)	Minimum slack (reported)
TT, 25°C, 1.80 V	18.68	0.35
SS, 100°C, 1.60 V	18.34	0.71
FF, −40°C, 1.95 V	18.80	0.22

Corner comparison graph

The graph compares the reported maximum and minimum slack across the three corners.



Note: slack units are not specified here. Positive slack in the listed summaries is encouraging, but complete timing sign-off requires reviewing all relevant setup/hold paths, constraints, unconstrained paths and the full reports.

6. Tools and Files

- Verilog HDL and Icarus Verilog/VVP for RTL simulation.
- GTKWave for waveform inspection.
- OpenLane and SKY130 PDK resources for the configured physical-design workflow.
- Mapped netlist, simulation VCD, synthesis log and corner-specific STA logs in the project files.

7. Results and Limitations

Functional simulation passed for the recorded valid and invalid password cases. Synthesis and three-corner STA reports are available. The available project notes previously identified an LVS mismatch involving the VGND and VPWR nets; therefore, this report does not claim that LVS passed. The final GDSII output and complete physical-design sign-off status must be verified against the actual run artifacts before submission.

8. Conclusion

The project demonstrates the implementation and verification of a password-controlled digital lock

using Verilog HDL. Simulation confirms the tested password cases, while synthesis and corner-based STA provide experience with the transition from RTL to implementation-oriented analysis. Further work should include resolving any remaining LVS issues, checking the final physical-design artifacts, and documenting the complete timing constraints and sign-off results.

References

1. Verilog RTL and testbench maintained in this project repository.
2. SKY130 PDK timing-library files used for the reported corner analyses.
3. OpenLane flow and the project's synthesis and STA reports.